Carbon-Intelligent Workload Scheduling for AI Data Centers

A Deterministic Optimization Approach to Joint Temporal and Spatial Scheduling

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Routing > Timing
Carbon savings come from *where* load runs, not *when*

Geography is the lever
Geographic flexibility dominates every other parameter by ∼ 10×

Large, robust savings
Up to **78%** carbon cut; **31%** under full operational limits

78%
max carbon cut
(vs. no optimization)

31%
under full
operational constraints

×10
geographic flexibility
over any parameter

5 / 2 yr
grids backtested
over 2024–2025

1 Motivation & Research Question

AI electricity demand is growing faster than grids can decarbonize, making the **operational carbon** of computing a first-order concern. Grid **carbon intensity (CI)** varies across **hours** and **regions**, so flexible load can move toward cleaner electricity. Existing schedulers are either **heuristic** (no optimality guarantee) or optimize the temporal and spatial dimensions **separately**.

*How can a deterministic model minimize carbon through **joint** temporal and spatial scheduling across regions, subject to realistic operational constraints?*

2 The Optimization Model

A linear program chooses the flexible load $x_{r,t}$ in each region r and hour t, driven by carbon intensity and the carbon-free energy fraction, to minimize operational carbon subject to seven operational constraints:

Linear program (solved exactly with HiGHS)

Decision variable: $x_{r,t} \geq 0$ — flexible load run in region r at hour t

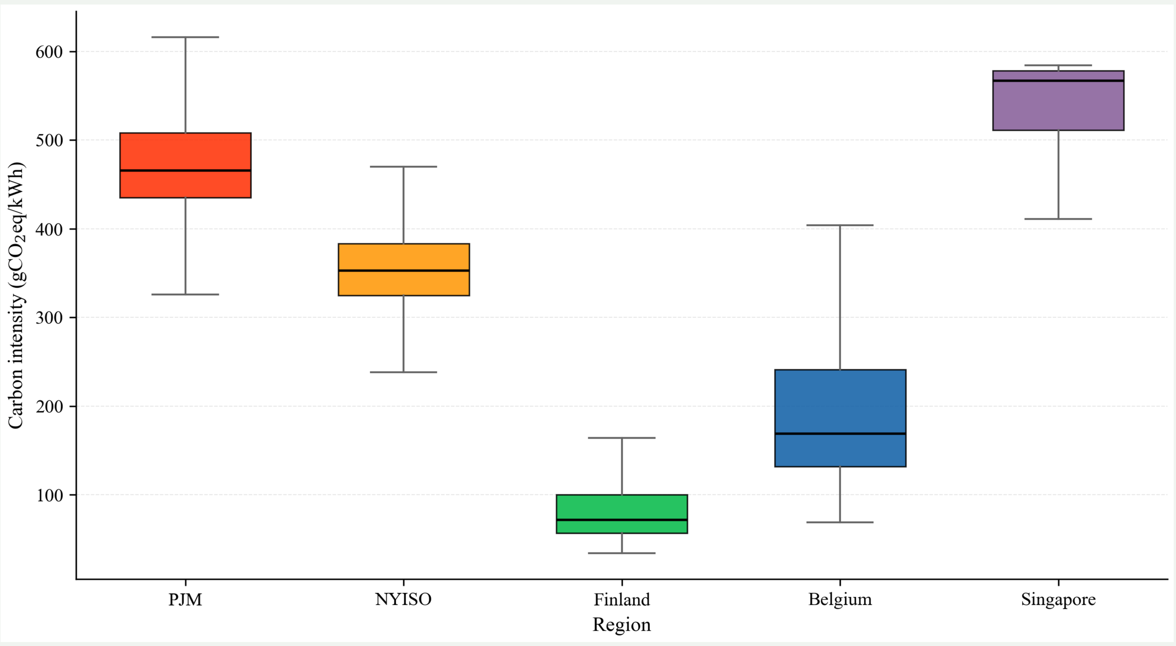
Objective: $\min_{x,M} \sum_{r,t} x_{r,t}\,\mathrm{CI}_{r,t}\,(1 - \alpha\,\mathrm{CFE}_{r,t}) + \eta M$

C1	C2	C3	C4	C5	C6	C7
Demand	Latency	Capacity	Geographic cap	Fairness	Ramp rate	Dynamic range

subject to seven operational constraints

3 Carbon Data: Five Grid Regions

Hourly CI and CFE from **ElectricityMaps**, **2024–2025** (17,544 h × 5 grids).



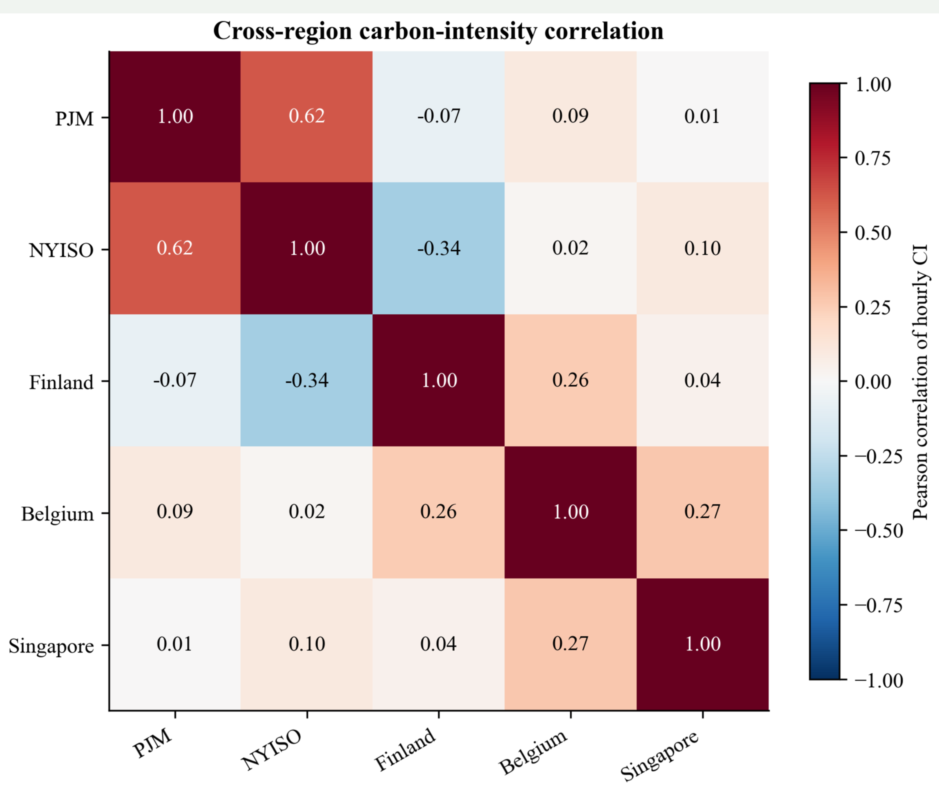
Median CI differs by almost **7×** (Finland ∼80 to Singapore ∼530 gCO_2eq/kWh).

Per-region descriptive statistics

Region	CI	sd	range	CFE	R^2
PJM	472	52	313–627	42.0	0.75
NYISO	354	44	215–499	51.1	0.77
Finland	83	35	34–285	94.0	0.97
Belgium	191	76	69–433	76.1	0.96
Singapore	533	60	391–584	2.7	0.06

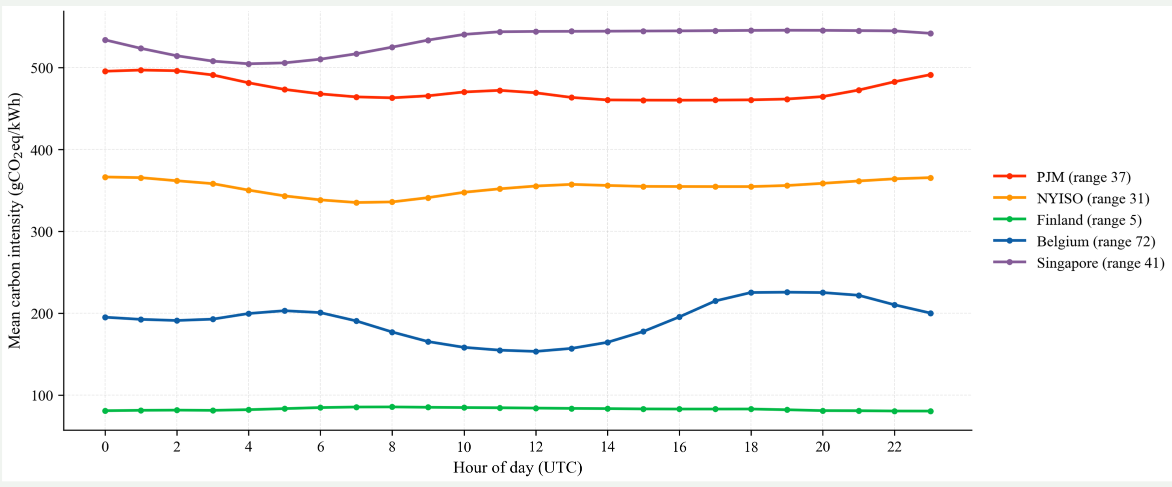
CI and its standard deviation and range in gCO_2eq/kWh; CFE in %; R^2 of CI on CFE.

4 Spatial Decorrelation Enables Routing



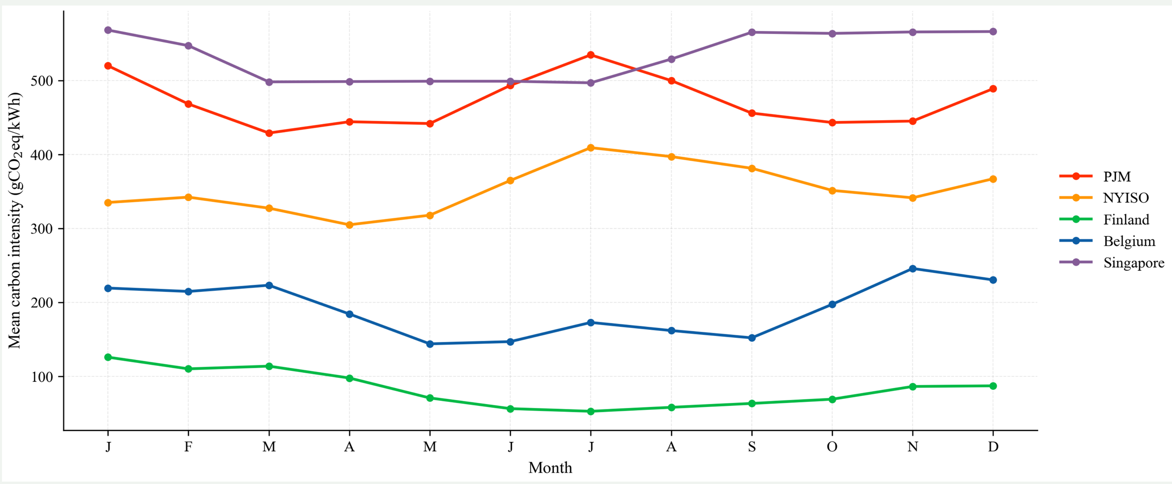
Off-diagonal CI correlations are near zero or negative: when one grid is dirty, another is usually clean — the structural precondition for spatial routing.

5 Diurnal Structure of Carbon



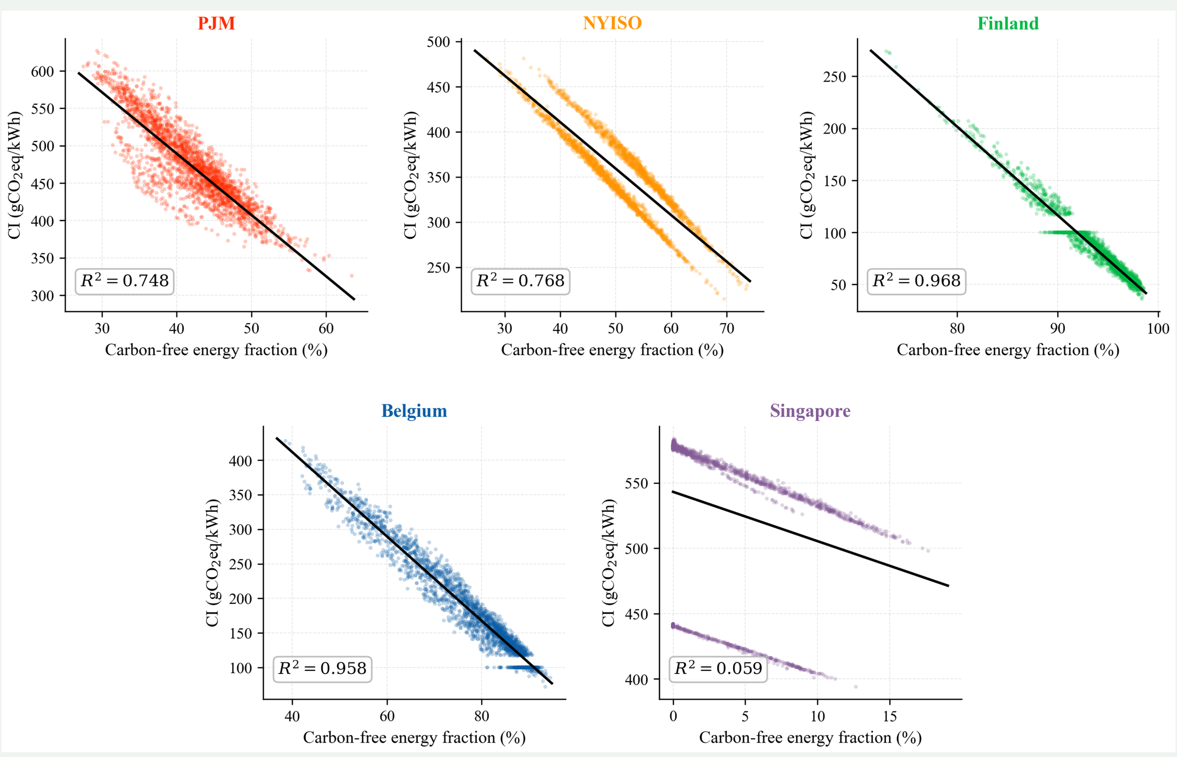
Intra-day CI range is largest in Belgium (midday solar) and smallest in Finland — yet Finland is the most variable region overall on a multi-day basis.

6 Seasonal Structure of Carbon



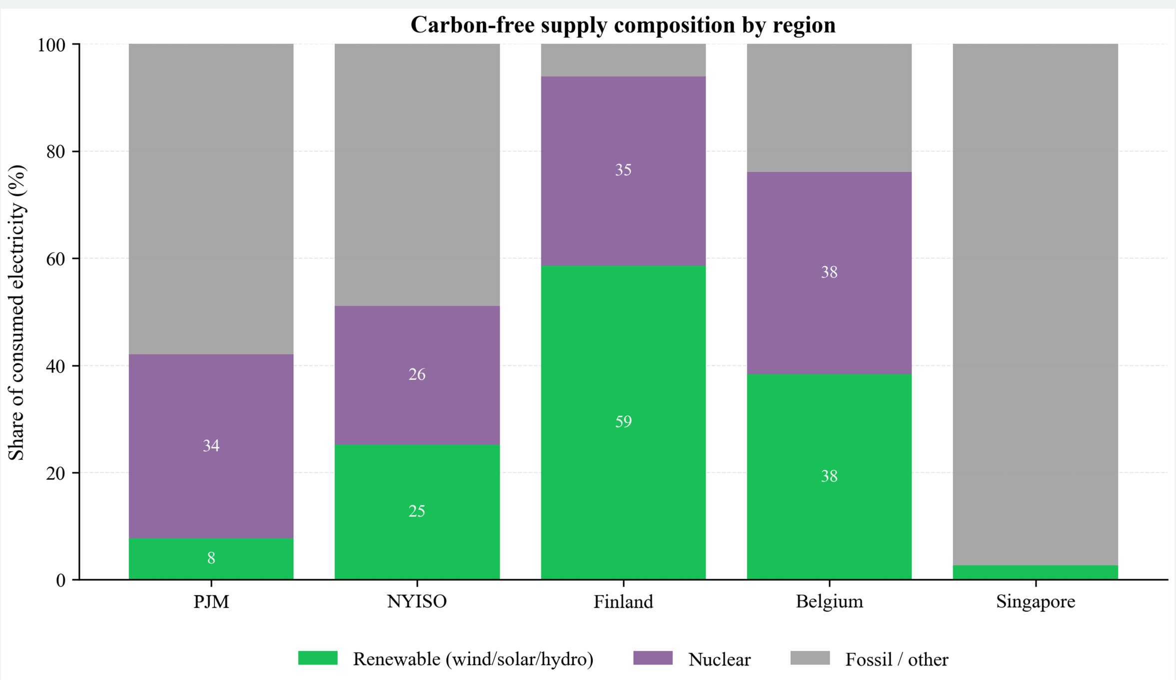
Finland and Belgium show seasonal troughs as renewables peak; PJM and Singapore stay high year-round. Seasonal swings exceed diurnal ones in clean grids, motivating the two-year horizon.

7 Why CFE? CI vs. Carbon-Free Fraction



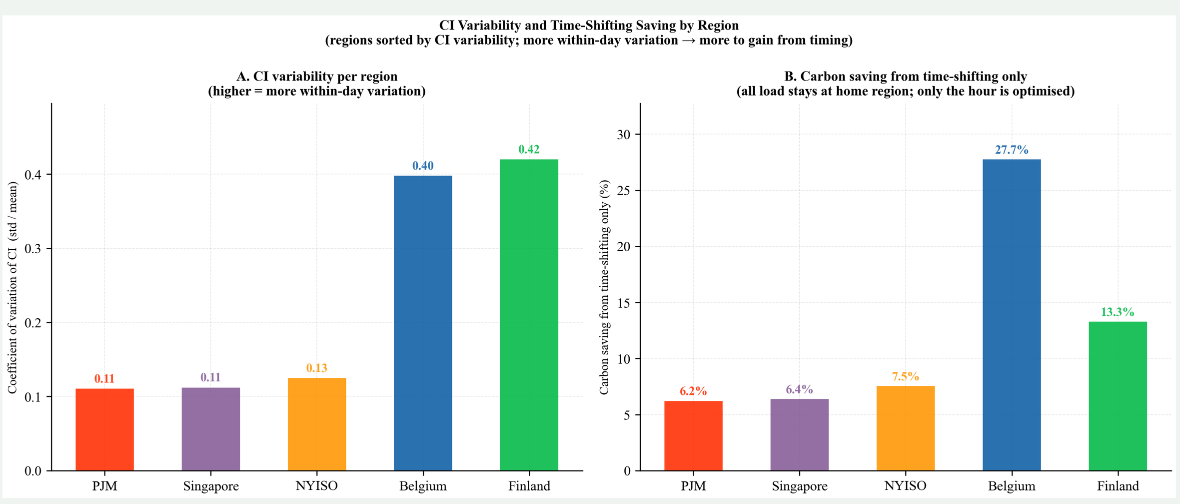
CFE explains most of the variation in CI (R^2 up to 0.97), except in gas-dominated Singapore — supporting CFE as a cleanliness signal alongside CI.

8 Carbon-Free Supply Composition



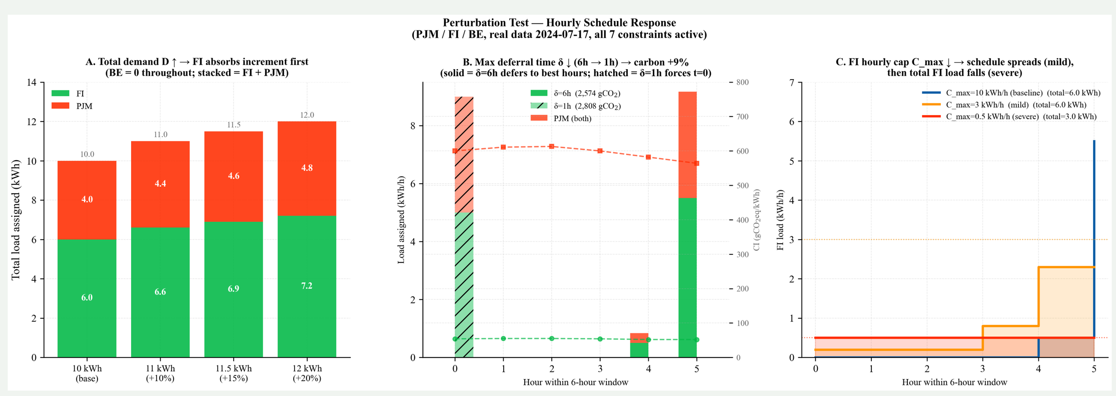
PJM and NYISO reach much of their carbon-free fraction through *nuclear*, not renewables — why a renewables-only signal would misrepresent them and CFE is used instead.

9 CFE Signal: Cross-Validation

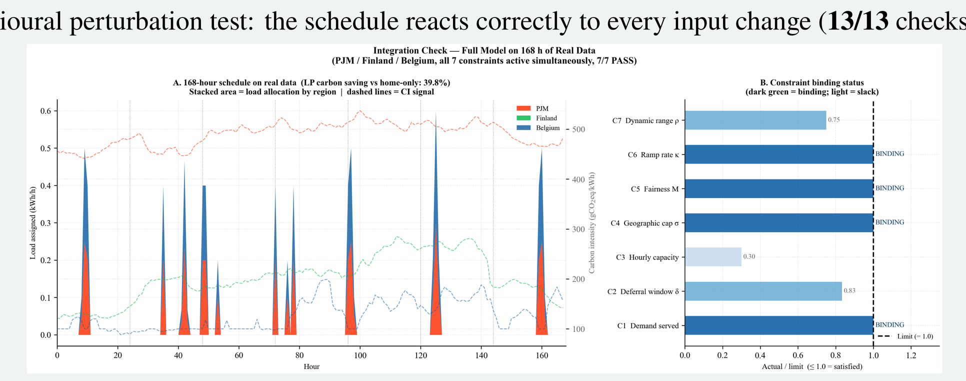


Five-fold cross-validation of the CI–CFE relationship confirms the signal generalises out-of-sample in every region except gas-fired Singapore.

10 Validation

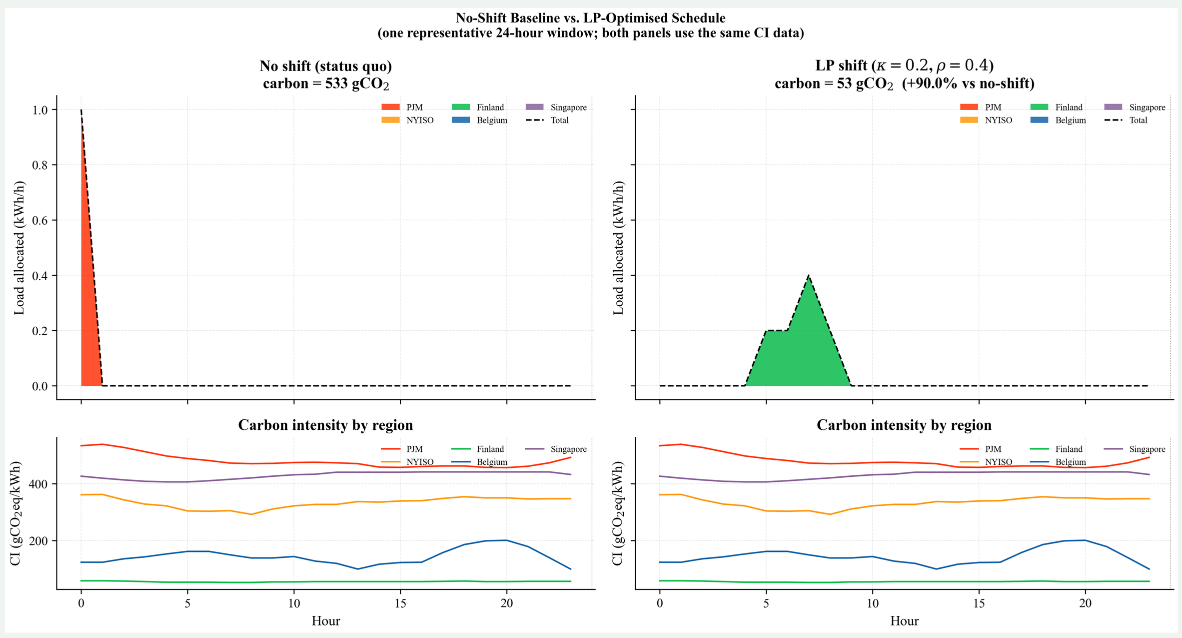


Behavioural perturbation test: the schedule reacts correctly to every input change (**13/13** checks pass).



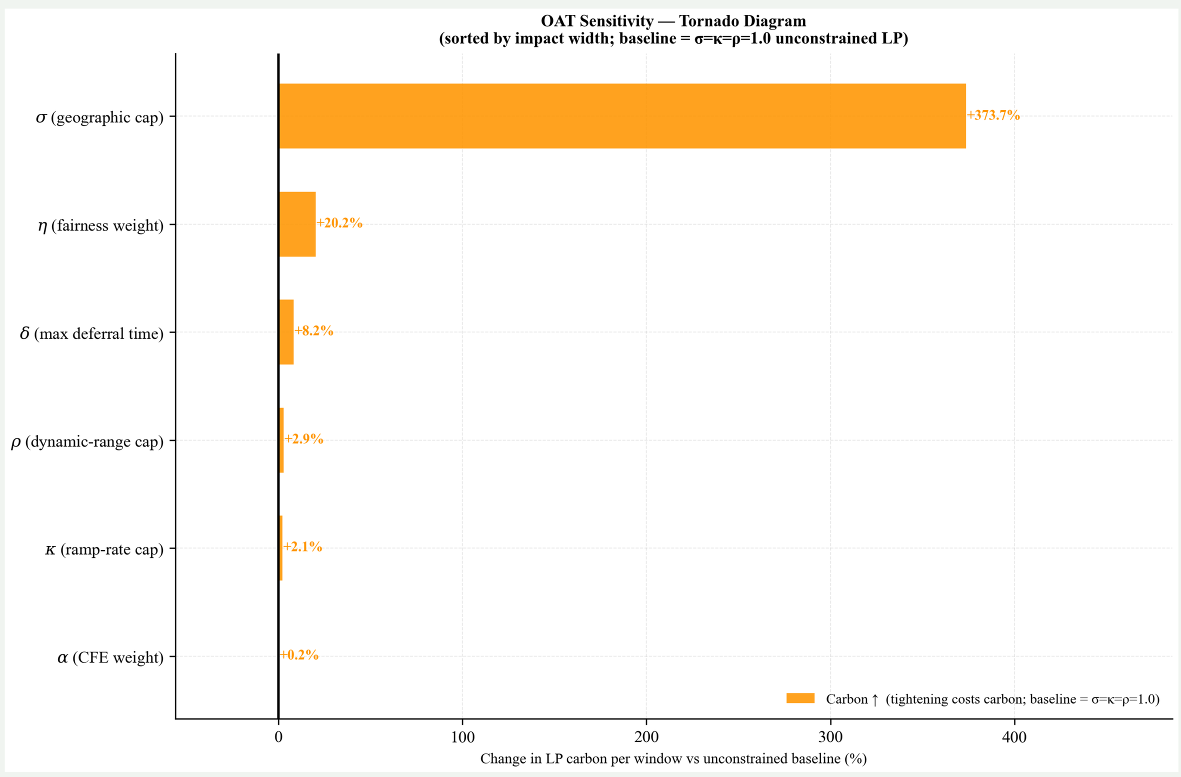
Integration check on 168 h of real data: all **7/7** constraints hold; LP cuts carbon **39.8%** vs. a home-only baseline.

11 Schedule Illustration



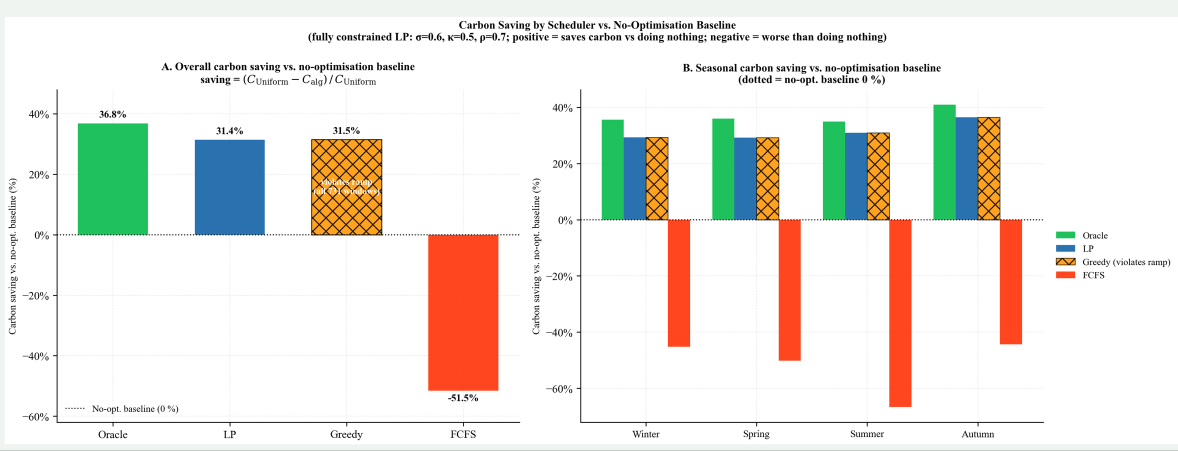
One window: the LP routes the batch to Finland and spreads it across hours to respect the ramp limit, cutting window carbon from 533 to 53 gCO_2.

12 Geographic Flexibility Dominates



One-at-a-time sensitivity. Tightening the geographic cap σ raises carbon by **+374%**, against **+20%** for equity η, **+8%** for the deferral window δ, and **<3%** for ρ, κ, α: how much load may move *across regions* sets the ceiling on savings, not how it is shaped in time.

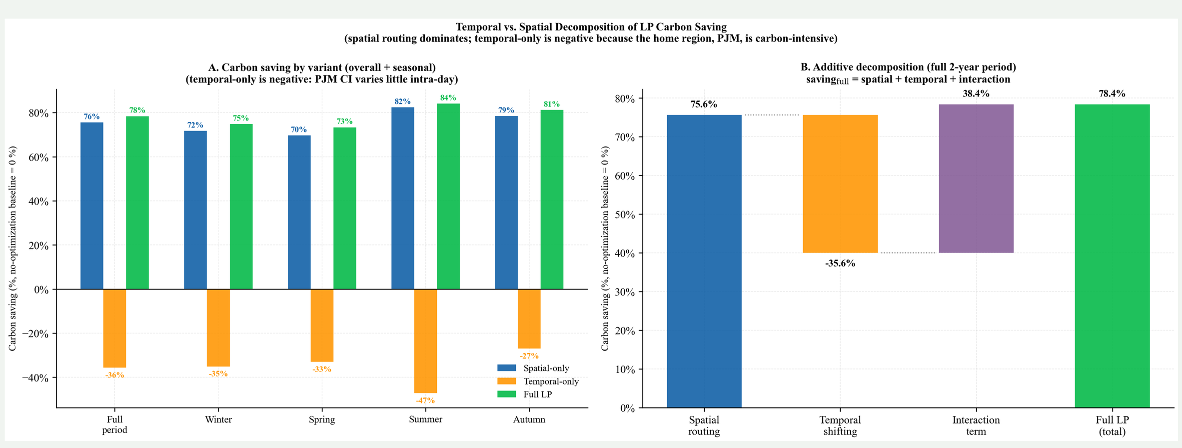
13 Comparison with Practical Schedulers



Method	Save	Feasible	Foresight
Oracle	36.8%	✓	week
LP (this work)	**31.4%**	✓	24 h
Greedy	31.5%	✗ (ramp)	24 h
FCFS	−51.5%	✗	none

The LP is the only method that attains near-ceiling savings *and* provably respects every constraint; greedy matches its carbon only by violating the ramp limit (all 731 windows).

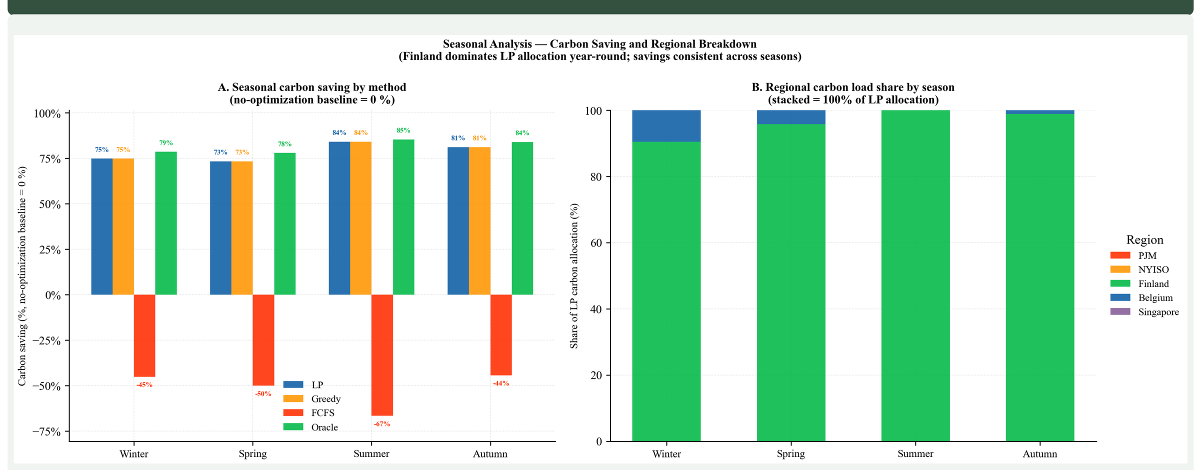
14 Where the Savings Come From



+75.6% routing −35.6% timing +38.4% interaction

Routing accounts for almost all of the saving; time-shifting *alone* can increase carbon. The two are strongly complementary.

15 Seasonal Robustness



Savings are stable across seasons (**73–84%**); Finland dominates the allocation year-round. The result is not an artefact of one season.

Future work: marginal emissions • forecast uncertainty • water co-optimization **Data:** ElectricityMaps 2024–2025 • **Solver:** HiGHS LP